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Power system transient stability assessment based on hierarchical graph pooling method considering missing data

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ABSTRACT

The transient stability assessment (TSA) model based on graph deep learning relies on complete system topology and its characteristics. However, electrical operation data may be missing during the measurement and transmission process, which can lead to a decline in model evaluation performance. To address this issue, this paper proposes a hierarchical graph pooling TSA method for power systems that accounts for missing data. First, the power system is modeled as a graph with topological connections, and the missing data is filled using the K-order neighborhood mean (KNM). Next, a masked graph autoencoder with jumping knowledge is developed to reconstruct the missing features. Finally, considering both the topological attributes of nodes and the temporal characteristics of electrical quantities, a well-designed graph pooling method is introduced. During hierarchical graph pooling, subgraph features of the power grid at different scales are extracted and fused to achieve accurate and reliable TSA. Test results on the IEEE 39-bus system and a provincial power system in China demonstrate that the proposed method can maintain high evaluation performance under various types of missing data, exhibiting strong robustness and practicality.

1. Introduction

Power system transient stability assessment is crucial to ensure that the power system can return to a stable operating state after a large disturbance such as a short-circuit fault. Currently, the power grid structure is undergoing a profound transformation driven by the large-scale integration of renewable energy sources and energy storage systems [1]. This shift is characterized by a transition from centralized, synchronous machine-based generation to a more decentralized, inverter-dominated architecture, resulting in increasingly complex and dynamic grid morphology. Such structural changes introduce new challenges to transient stability, as the system exhibits reduced inherent inertia and greater operational uncertainty. In response, significant research efforts are being dedicated to enhancing power system transient stability through improved system operation strategies, advanced control technologies, optimized resource dispatch, and enhanced management mechanisms. These include the development of adaptive protection schemes [2], grid-forming inverter controls [3], wide-area monitoring and control systems [4] to support dynamic performance.

Given the growing complexity and stochastic nature of modern power systems, the ability to perform fast and accurate transient stability assessment has become increasingly critical. Such assessment not only facilitates the timely deployment of emergency control measures but also provides the foundational basis for ensuring the secure and stable operation of the grid under increasingly dynamic conditions. The traditional TSA method using time domain simulation (TDS) [5] solves differential–algebraic equations, which is time-consuming. Similarly, the energy function method [6] has limitations in large power systems due to the complexity of finding energy functions and high computational demands.

Phasor measurement unit (PMU) [7] is a high-speed digital system that can collect real-time data of electrical quantities such as voltage and current in various parts of the power system at a frequency of hundreds of times per second, and send these data to the monitoring center for processing and analysis through a wide-area communication network. These data can be used to construct the state vector of the power system and then applied to proposed scheme. With the advancement of artificial intelligence technology and the widespread adoption of PMU in wide-area measurement systems, researches on data-driven TSA have advanced rapidly and show great promising for achieving more reliable and efficient power system operation [8,9]. By leveraging large volumes of monitoring data, machine learning algorithms can automatically

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learn patterns and relationships between system variables to generate accurate predictions. Various machine learning models have been used in power systems for stability prediction such as decision trees (DT) [10], support vector machines (SVM) [11], artificial neural network (ANN) [12]. Meanwhile, deep learning techniques such as convolutional neural network (CNN) [13–15], stacked auto-encoder (SAE) [16], long short-term memory (LSTM) [17] can extract high-level features from raw data and capture complex nonlinear dynamics, thereby enhancing the generalization ability of the model.

However, these methods mainly rely on feature extraction and classifier construction based on state variables, rarely consider the influence of topology on system stability, and perform poorly when the system topology changes. Graph neural network (GNN) [18,19] provides a new technical route for considering topological correlation in high-dimensional feature aggregation. It can regard the buses in the power system as nodes and the branches as edges. By integrating grid topology information into the neural network, features representing both nodes and graphs can be extracted through multi-layer convolution, pooling, and other operations, thereby enabling the modeling of the power system's structure and state [20–25].

Although existing power system TSA methods based on graph deep learning have made progress in extracting power grid topology information [24,26–28], these approaches assume that PMU measurement data is complete and that full power system topology and characteristics are available when building graph deep learning models. However, in practice, PMU data is often subject to loss and abnormalities. According to [29], the reliability of the wide-area measurement system for the 14-bus system is approximately 97.35 %, with a data loss probability of 2.65 %. Similarly, in the daily operation of China's power grid, around 10 % to 30 % of PMU measurements exhibit outliers or missing values [30]. This data incompleteness can degrade the performance of GNN-based TSA models [31]. Additionally, existing research fails to account for the differences in the electrical characteristics of individual nodes when establishing the GNN model, and does not effectively model the information transmission characteristics between multi-scale subgraphs in the grid topology. The electrical quantities at each node exhibit both similarities and differences. Extracting and integrating subgraph features from different nodes, based on their time series trajectories and topological structure, can enhance the performance of the TSA model.

Considering that existing research does not adequately address the issue of missing data and lacks effective modeling of the information transmission characteristics between multi-scale subgraphs of the power grid, a two-stage TSA method is proposed. In the first stage, a masked graph autoencoder with jumping knowledge is employed, based on KMN filling, to reconstruct the missing features. In the second stage, using the reconstructed complete power system data, the characteristics of the topological subgraphs at different scales are fully explored through hierarchical pooling. This allows for the extraction of structural and feature information from the diagram and enables the discrimination between the original input graph and the output stability result. Overall, the main contributions of this work are as follows:

1) A graph-based missing data imputation method is developed based on the topological relationships of the power grid. It integrates KNM filling method and introduces jumping knowledge into a masked graph autoencoder, enabling high-precision data reconstruction.
2) The method can handle various PMU data missing scenarios, allowing for direct processing of historical PMU measurements from the actual power grid for training TSA models.
3) During hierarchical graph pooling in the TSA, node features are selected based on node importance evaluations. The feature information of unselected nodes is not discarded; instead, graph attention network (GAT) is used for first-order neighborhood aggregation, so as to realize the information extraction of network subgraphs at multiple scales and enhance the global feature extraction ability.

The remainder of this paper is organized as follows. Section 2 introduces the process of PMU missing data filling and graph data reconstruction. Section 3 outlines the hierarchical graph pooling TSA scheme based on reconstructed data. Section 4 demonstrates the application of the proposed method in two different systems and various scenarios. The conclusions are presented in Section 5.

2. Missing data imputation and graph data reconstruction

In this section, a graph data reconstruction method is proposed to impute missing data in the power system PMU measurement. First, the types of missing data are described. Then, the missing data is filled using the neighborhood mean method, initially based on topological connections. Finally, the data is reconstructed utilizing a graph autoencoder.

2.1. Types of PMU missing data and mask setting

In actual power systems, factors such as equipment failure, communication congestion, and other issues can lead to missing or abnormal PMU data. This reduces the observability and monitoring capabilities of the power system, compromising the accuracy of state evaluations and seriously affecting system stability. As shown in Fig. 1, the transient sample data from PMU measurements is typically represented as a two-dimensional array and the mask matrix indicate the observability of the power grid under different PMU data missing conditions. Based on the type of missing PMU data and its distribution within the sample, it can be classified into two categories, and masks are configured for the complete observation data respectively.

1) Single point random missing: most PMU missing data are randomly distributed within a specific value range in the sample, typically due to transmission noise or communication failure. Given a complete observation sample $X \in \mathbb{R}^{v\times d}$, which consists of V nodes, each node has an associated attribute matrix with d features. To represent the missing data positions, the mask matrix M has the same shape as X. Its values are binary: 0 indicates that the corresponding value in X is missing, while 1 indicates that the value is present. Since the missing values are randomly distributed, the positions of the missing data are independent of each other.

2) Consecutive multi-point missing: consecutive multi-point missing data typically result from prolonged PMU failures and are distributed in a regular pattern over a continuous range of values within the sample. To represent this type of missing data, a mask matrix is created by applying a fixed-length block of missing values to the feature sequence of each node. In extreme cases, one or more complete sequences may be missing in the transient sample due to a PMU fault during the observation period or incomplete PMU configuration.

2.2. Missing data imputation with KNM filling

Unlike traditional missing data imputation methods [32–34], the proposed approach operates on graph-structured data, considering the topological properties of power systems. The power system is represented as a graph, where nodes correspond to generators, transformers, and other equipment, and edges represent their connections. For each graph, the input consists of the characteristics of each node and the relationships between them. The input two-dimensional matrix data are then mapped to the corresponding nodes as node features. The proposed imputation method, based on KNM filling, uses the adjacency relationship of nodes to fill missing features. Specifically, the missing features are filled through the following steps:

Initially, the input data consists of the edge index, node feature matrix, and the mask of the graph. In the first step of data processing, the positions where the mask is 0 are set to zero, effectively assigning missing features a value for subsequent calculations:

$$\widetilde{X} = X \odot M \tag{1}$$

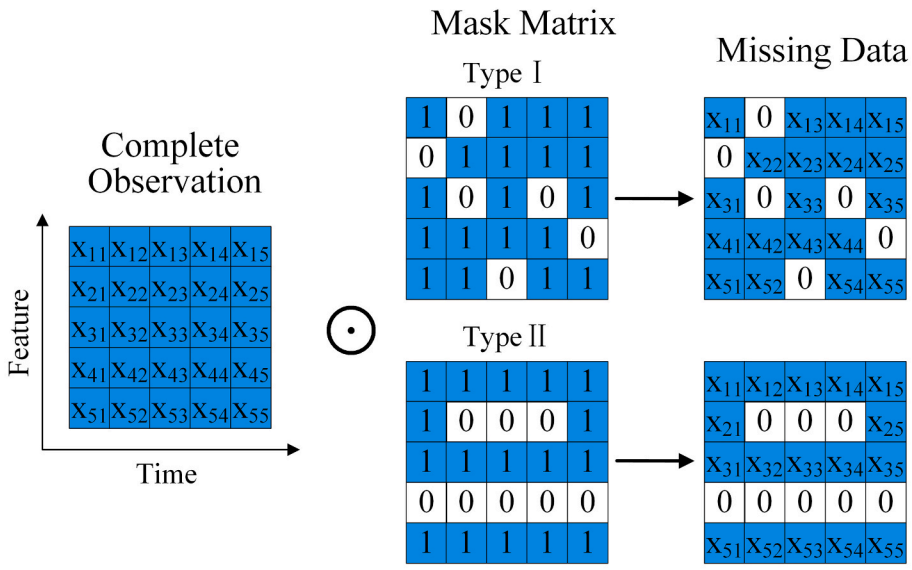


Fig. 1. PMU missing data types and mask settings.

where $\odot$ denotes the elementwise multiplication. The degree of each node is then calculated through sparse matrix multiplication [35] and denoted as D.

Subsequently, the neighbor information is used to fill in the missing node features. To achieve this, sparse matrix multiplication is applied again to calculate the mean of the neighbor features for each node. Specifically, the weighted sum of the features of its adjacent nodes is divided by the degree of the node:

$$\overline{X} = \left(A' \cdot \widetilde{X}\right) / D \tag{2}$$

where A' is the modification of original adjacency matrix A. For the topological connections in the power system, when PMU data is missing, the number of first-order neighbors for some nodes may decrease, resulting in insufficient aggregation information to accurately capture the spatial characteristics of the power grid. To address this, the k-order neighborhood of the target node is considered by adding the higher-order adjacency matrix to the original adjacency matrix, as shown in (3). Meanwhile, as the order increases, the influence of each higher-order neighbor on the central node diminishes, and the weight of the k-order adjacency matrix is reduced, as shown in (4):

$$A' = A + \alpha_2 A^2 + \alpha_3 A^3 + \ldots + \alpha_k A^k \tag{3}$$

$$\alpha_k = e^{-(k-1)} \tag{4}$$

Finally, the completed values for each node at the missing feature positions are returned, corresponding to the mean of its neighbor features. This method leverages the graph structure and neighbor information to effectively fill in the missing features. However, this simple interpolation approach may not perform well in power grids with high measurement noise. To further enhance the accuracy of missing data interpolation, a graph encoder module is designed to learn and estimate the interpolation error between the data after KNM filling and the actual measured values.

2.3. Graph data reconstruction with masked graph autoencoder

The graph data imputation module is used to further achieve high-precision missing data reconstruction. As shown in Fig. 2, for the given graph G, the goal is to learn its encoding to produce the node embedding h_G, which is then optimized to handle measurements with missing inputs. The node embedding is learned by graph isomorphism network convolution (GINConv) [36], which treats the graph convolution operation as an aggregation process of node features, and updates the representation of nodes by aggregating the features of their neighboring nodes. For each layer of GINConv, information is collected from each neighboring node as well as the node itself through a series of nonlinear transformations and weighted summation operations. This process updates the hidden state of node v. The hidden state feature of the node at the ith layer is denoted as $h_v^{(i)}$.For each node, the hidden state at the $(i+1)$th layer is calculated as:

$$h_v^{(i+1)} = \mathrm{MLP}^{(i)}\left((1+\varepsilon^i)h_v^{(i)} + \sum_{u \in N(v)} h_u^{(i)}\right) \tag{5}$$

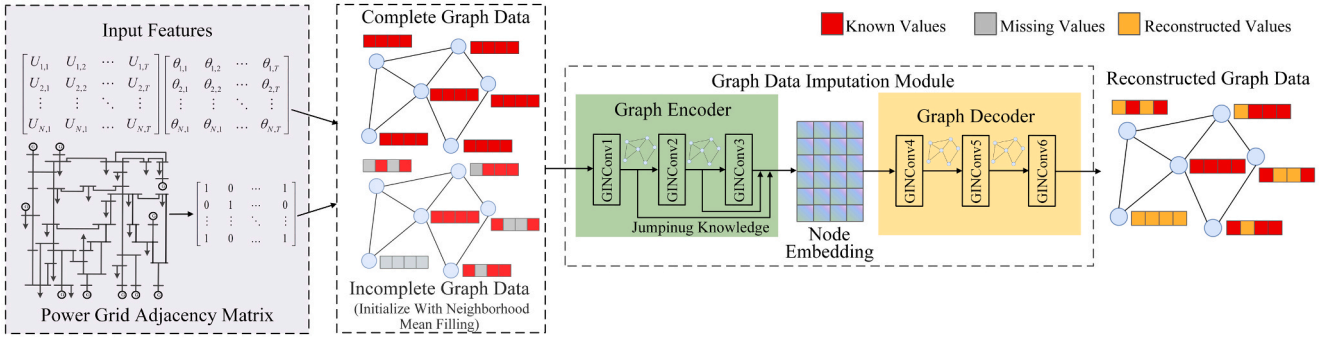


Fig. 2. Graph data reconstruction based on masked graph autoencoder.

where $MLP^{(i)}$ denotes a multilayer perceptron that transforms the input state, $N_{(v)}$ represents the neighbor set of node v, ε^i is a learnable parameter and initialized to 0, which represents the modification of the node's own characteristics at the ith layer, so that the feature vector of the node is not affected by the order of the neighbor nodes.

In the GNN structure, the node representation at each layer is updated progressively based on the representations of its neighboring nodes. However, in multi-layer networks, this process can lead to a gradual loss of local details in the node representations. To address the challenge of excessive reliance on the graph's structure, particularly in capturing long-range dependencies between distant nodes, jumping knowledge [37] is introduced into the graph encoder. This approach combines node representations from different layers through jump connections, rather than relying solely on the information from the final layer.

The node embeddings of the last layer are then fed into the graph decoder, which is also contains several GINConv layers. The decoder maps h_G back to the original space to reconstruct the missing features and generate the output X'. The final imputed data X_{rec} is a combination of the reconstructed data, based on the locations of the missing values, and the original observed data, which will be used as input for the TSA model:

$$X_{rec} = X \odot M + (1 - M) \odot X' \tag{6}$$

3. Hierarchical graph pooling model for TSA

3.1. Proposed TSA model architecture

The overall architecture of the proposed hierarchical graph pooling method is shown in the Fig. 3. The feature extractor comprises three blocks, each of which consists of a graph convolution layer and a graph pooling layer. Initially, the topological structure of the power grid is mapped to a graphical representation, ensuring that the node number and connection relationships of graph data correspond to original power grid, the input to the model is the reconstructed missing data and connection relation matrix of each node in the graph. Subsequently, the output of each block is summarized in the readout layer to capture the high-order characteristics of the power system subgraphs at different scales. Then, the outputs from the three blocks are summed to produce a fused graph representation vector, which incorporates multi-scale subgraph features. This vector is then passed to a classifier, consisting of several fully connected layers, for classification. Finally, the classifier outputs the transient stability prediction labels.

The graph neural network for feature extraction still uses GINConv, with the pooling operation combined with the convolutional operation to reduce the dimensionality of the representation and obtain a compact representation of the entire graph. Next, the operation of the pooling layer will be introduced in detail.

3.2. Specific operation of graph pooling layer

When pooling node representations in graph data, it is essential to take into account both the distinctive local structural information and the characteristic information of each node. Through this dual perspective, the significance of nodes is assessed, and nodes with the highest scores are chosen for feature fusion, aligning with the objective of pooling. The specific functionality of the self-adaptive pooling (SAGpool) layer is visually depicted in Fig. 4.

3.2.1. Rating the importance of topological structures

The graph convolutional network (GCN) [38] utilizes the adjacency and degree matrices as they offer an effective means to characterize the relationships between nodes and their local topology, such as node degree. Drawing from the spectral method in graph signal processing, this concept transforms the signal processing task on the graph into operations within the graph's spectral domain. Through normalization and transformation of the adjacency and degree matrices, they can be interpreted as the Laplacian matrix of the graph. This approach is not only simple and efficient but also retains global and local topological structure information, therefor it is suitable for evaluating the importance of nodes. The calculation of the structural information score for a node using GCN is as follows:

$$S_1 = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}HW\right) \tag{7}$$

where $\widetilde{A}$ is the adjacency matrix after adding self-connection, $\widetilde{D}$ is diagonal node degree matrix, H is the feature matrix and W is the weight vector. σ is the activate function and here tanh function is used to limit the score result to [-1,1].

3.2.2. Scoring of temporal feature information

Gated recurrent unit (GRU) [39] is a recurrent neural network model that can model sequence data and retain sequence information in the sequence. The GRU consists of two gates (reset gate and update gate) and a memory unit. The calculation process can be expressed from (8) to (12):

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \tag{8}$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \tag{9}$$

$$\widetilde{h_t} = \tanh(W_h x_t + r_t \odot h_{t-1} + b_h) \tag{10}$$

$$h_t = (1 - z_t) \odot U_h h_{t-1} + z_t \odot \widetilde{h_t} \tag{11}$$

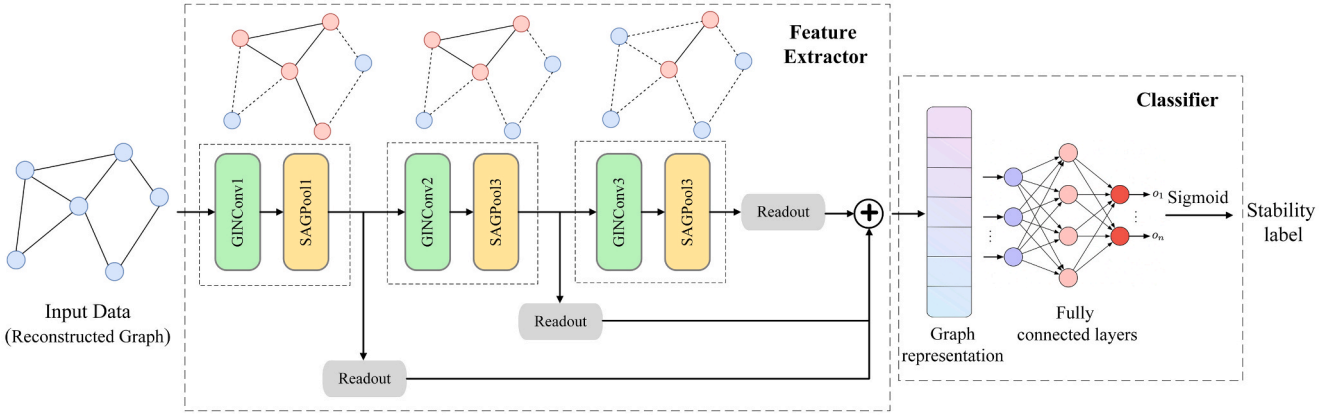


Fig. 3. Overall structure of the proposed hierarchical graph pooling model for TSA.

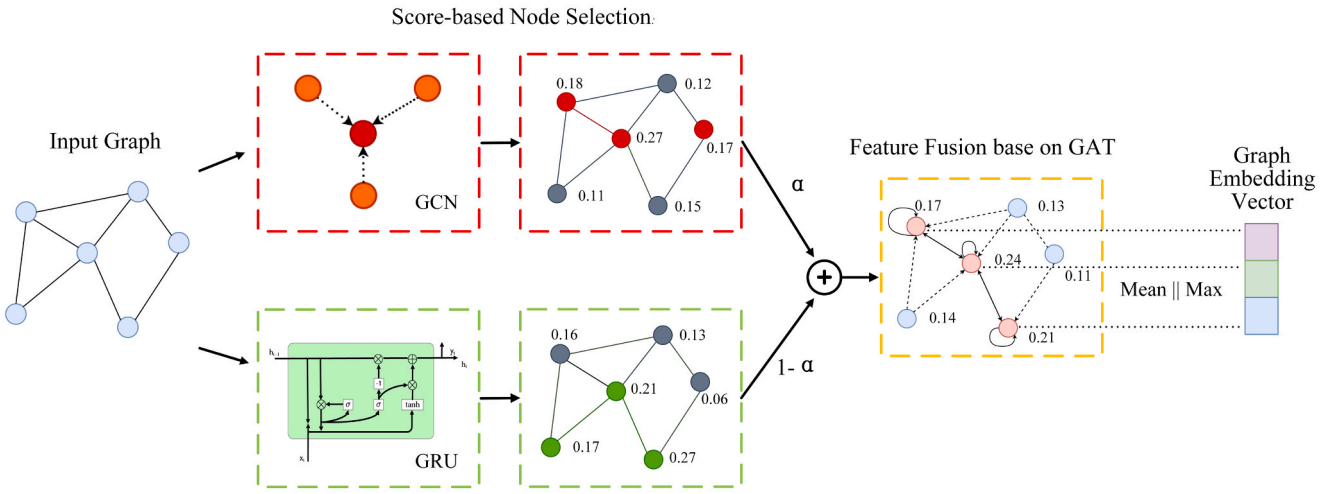


Fig. 4. The specific operation of pooling layer.

$$y_t = \sigma(W_y h_t + b_y) \tag{12}$$

where x_t is the input vector at time step t, h_t is the state vector which summarizes the information from the first t inputs, r_t is the reset gate vector, z_t is the update gate vector, $\widetilde{h}_t$ is the candidate state vector, W_r, Wz, W_h, W_y, U_r, U_z, U_h, b_r, b_z, b_h and b_y are learnable weight matrices and bias vectors, $\odot$ represents element-wise multiplication. The reset gate r_t controls how much the previous state vector h_{t-1} influences the current candidate state vector $\widetilde{h}_t$. The update gate z_t controls how much the past information contributes to the current state. The candidate state vector $\widetilde{h}_t$ is obtained by combining the current input vector and the previous state vector with the reset gate r_t. The final state vector h_t is a weighted average of the previous state vector h_{t-1} and the candidate state vector $\widetilde{h}_t$, with weights determined by the update gate z_t.

When analyzing the importance of nodes when power system failure occurred, it is significant to consider changes in voltage and phase angle over time. To capture these temporal characteristics, GRU can be used to extract node temporal information. The expression for using GRU to extract node timing characteristics would involve defining the input sequence for each node, which would include voltage and phase angle measurements taken at a specified time period. These sequences would then be fed into the GRU, which would apply its gating mechanisms and memory unit to aggregate the feature information across the entire sequence. The output of the GRU for each node would represent a summary of the node's temporal characteristics, capturing how its voltage and phase angle changed over time after the system failure and we define it as time characteristics importance score:

$$S_2 = \sigma(\mathrm{GRU}(H)) \tag{13}$$

3.2.3. Score-based node selection

Considering the topological importance and time characteristics influence of each node, these two scoring methods are combined to objectively select nodes.:

$$S = \alpha S_1 + (1-\alpha) S_2 \tag{14}$$

where the weight α is a hyperparameter. According to the ranking results, top m nodes are selected as the result of pooling and m is also a hyperparameter.

3.2.4. Feature fusion mechanism

In order to obtain a more efficient graph representation and make use of the feature information from unselected nodes, first-order neighborhood aggregation will be conducted based on the results of node selection. This operation involves using GAT to aggregate the feature information from neighboring nodes:

$$h_i^{(l+1)} = \sigma\left(\sum_{j\in N_i} a_{ij}^{(l)} W^{(l)} h_j^{(l)}\right) \tag{15}$$

$$\alpha_{ij}^{(l)} = \frac{\exp\left(\mathrm{LeakyReLU}\left(\vec{a}^{(l)}\right]W^{(l)}h_i^{(l)} \Big\| W^{(l)}h_j^{(l)}\right])\right)}{\sum_{k\in N(i)}\exp\left(\mathrm{LeakyReLU}\left(\vec{a}^{(l)}\right]W^{(l)}h_i^{(l)} \Big\| W^{(l)}h_k^{(l)}\right])\right)} \tag{16}$$

where $W^{(l)}$ is the lth layer weights, $a_{ij}^{(l)}$ is the lth layer normalized coefficient between node i and node j.$\|$ represents concatenation. LeakyReLU$(\cdot)$[40] denotes an activation function. For each selected node, its neighbor nodes in the graph are identified and their features are extracted. These features would be passed through a GAT layer, which would apply attention weights to each neighbor node based on their relevance to the selected node. The output of the GAT layer would be a weighted sum of the neighbor node features, with weights determined by the attention mechanism.

After performing first-order neighborhood aggregation, a comprehensive representation of the graph can be obtained, incorporating both selected and unselected nodes. By employing a pooling operation, the individual node-level features are transformed into a unified graph-level feature vector known as readout. Maximum pooling selects the largest hidden state vector among all nodes, while average pooling calculates the average value of the hidden state vectors of all nodes as the hidden state vector of the whole graph. The final graph embedding vector s is obtained by concatenating the results of the maximum pooling and average pooling operations. The calculation formula is expressed as follows:

$$s = \left(\frac{1}{V}\sum_{i\in V} h_i\right) \Big\| \left(\max_{i\in V} h_i\right) \tag{17}$$

Hierarchical pooling aggregates node features in the graph at various levels to create a higher-level representation, effectively capturing the topology and feature information of the complex power system graph. The graph embedding vector, obtained by summing each graph representation, will serve as the input for the classification task.

3.3. Overall process of TSA

The comprehensive flow of the TSA method based on missing data reconstruction, as illustrated in Fig. 5, consists of two parts: offline model training and online application.

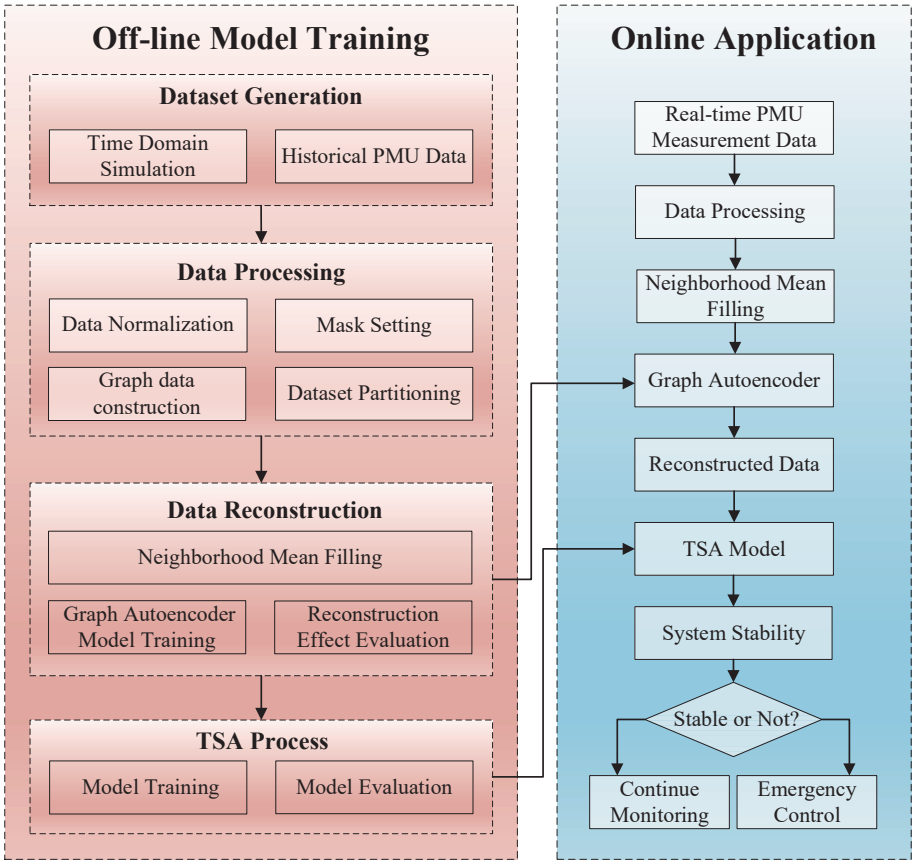


Fig. 5. The overall framework of the TSA Off-line model training.

3.3.1. Offline model training

Firstly, the dataset is generated and preprocessed. The input data includes the electrical measurement parameters of each bus node and the topology of the power grid. The electrical parameters of the nodes are standardized and formatted into a feature specification suitable for model input, while the original adjacency matrix is constructed based on the grid structure topology. Each sample in the dataset is labeled based on its transient stability index (TSI) [41,42], which is expressed as:

$$I_{TSI} = \frac{360^{\circ} - |\Delta\delta_{\max}|}{360^{\circ} + |\Delta\delta_{\max}|} \tag{18}$$

where $\Delta\delta_{\max}$ is the maximum value of the power angle difference between any two generators after the fault. When $I_{TSI} > 0$, it is considered that the system is stable, otherwise it is considered that the system is unstable. The stable sample is marked as [1,0], and the unstable sample is marked as [0,1].

All samples are then divided into a training set and a test set based on a predefined proportion. After applying the mask matrix, the data with missing values are generated. Initially, the neighborhood mean algorithm is used to impute the missing values, which are subsequently fed into the graph autoencoder model. The learning objective is to minimize the difference between the reconstructed and original values:

$$L_{rec} = \|X - X_{rec}\|^2 \tag{19}$$

where $\|\cdot\|$ is the Euclidean distance. The performance of the model is measured by calculating the mean absolute percentage error (MAPE) and mean square percentage error (MSPE) of missing data. with the calculation formulas given as follows:

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\hat{x}_i - x_i}{\hat{x}_i}\right| \tag{20}$$

$$\text{MSPE} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\hat{x}_i - x_i}{\hat{x}_i}\right)^2 \tag{21}$$

where n is the number of missing data, and $\hat{x}_i$ and x_i represent the reconstructed and true values, respectively. The reconstructed graph data is used as input of the TSA model, and the binary cross entropy loss function measures the gap between the output of the model and the real label, with the goal of optimizing the network parameters:

$$L_{cls} = \frac{1}{N}\sum_{i=1}^{N}[y_i \log f_i + (1 - y_i)\log(1 - f_i)] \tag{22}$$

where N is the number of samples. The true label of sample i is y_i and f_i is the predicted value of the model. Based on the confusion matrix of the Table 1, the performance of the TSA model is measured by four indicators: accuracy A_{cc}, recall R_{ec}, precision P_{re} and F1 score F_1. The calculation expression of these metrics are explain in (23) to (26):

$$A_{cc} = \frac{N_{\text{TP}} + N_{\text{TN}}}{N_{\text{TP}} + N_{\text{FP}} + N_{\text{TN}} + N_{\text{FN}}} \times 100\% \tag{23}$$

$$R_{ec} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \times 100\% \tag{24}$$

$$P_{re} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \times 100\% \tag{25}$$

Table 1
Confusion matrix.

Label	Predicted as stable	Predicted as unstable
Stable	TP	FN
Unstable	FP	TN

$$F_1 = \frac{2 \times P_{re} \times R_{ec}}{P_{re} + R_{ec}} \times 100\% \quad (26)$$

where A_{cc} is the primary metric used to measure and improve the performance of the model. R_{ec} represents the proportion of correct classification in all stable samples. P_{re} is the proportion of actual stable in all samples classified as stable and F_1 denotes the harmonic mean of the recall rate and the precision rate.

3.3.2. Online application

The online deployment of the trained model hinges on high-precision and high-speed PMU. PMUs measure multiple parameters within the power system, and the measurement outcomes are transmitted to the data processing system via various communication protocols for analysis and processing. Under conditions of missing measurement data, the observable node features are processed to obtain the reconstructed data, which is then input into the TSA model to quickly obtain the system's stability results. Emergency control devices do not activate immediately following a line fault, or protection action. Instead, the system's stability is first evaluated. If the evaluation result is stable, the emergency control device remains inactive and the system can be monitored continuously. However, if unstable is detected, appropriate control measures, such as load shedding or generator trip, are implemented. The effectiveness of these control measures increases when applied earlier.

4. Case study

The proposed method is implemented on two different power systems: The IEEE 39-bus system and a provincial power grid in China. The database is generated based on TDS in Power System Analysis Software Package (PSASP). The data reconstruction model and TSA model is developed on Pytorch. All the tests are fulfilled on a computer with Intel Core i7-12700H@ 2.30 GHz CPU,16 GB RAM and RTX 3060 Laptop GPU.

4.1. Data generation and model setting

The IEEE 39-bus system is a regional power grid in New England, North America, with 10 generators, 46 transmission lines and 12 transformers. The expected fault conditions include varying load levels under both topology and the "N-1" topology (ranging from 80 % to 120 % in 5 % steps), along with adjustments to the generator output to maintain system power balance. Three-phase short-circuit faults with durations of 0.1 s, 0.2 s, and 0.3 s occur at different positions along each line (10 % to 90 % from the head end in 20 % steps). Finally, 19,335 samples are obtained with 13,081 stable ones and 6254 unstable ones. Each sample contains the input characteristics, the topological connection relationship and the stability label of the system. The system sampling frequency is 120 Hz, and the input features consist of the voltage and phase angle trajectories of 12 consecutive points at each node, recorded after fault removal.

To simulate noise interference in a real-world measurement environment, Gaussian noise with a mean of 0 and a variance of 0.01 pu is added to all sample features. The sample set is divided into training set and test set according to the ratio of 8: 2. For the training dataset, a 20 % random data loss rate and 10 % node feature loss are introduced to simulate PMU failure and data loss [43].

The Adam optimizer [44] is used to train the graph autoencoder with a learning rate of 1e-4. The dimension of the node embedding is 64. Both the graph encoder and decoder consist of three GINConv layers, each containing an MLP with two connection layers and 64 neurons. The ReLU activation function and batch normalization [45] are applied after each GINConv layer to improve training stability.

The learning rate for TSA model training is set to 1e-3, and the Adam optimizer is used as well. The configuration of the GINConv layer remains the same as described above. The parameters of SAGpool are determined by comparing different configurations and selecting the one that yields the best model performance. Ultimately, the pooling ratio is set to 0.5, and the score weight α is 0.6. Table 2 provides detailed structural information for the proposed hierarchical graph pooling model.

4.2. Data generation and model setting

To demonstrate the superiority of the proposed algorithm, it is compared with the K-nearest neighbor (KNN) interpolation method, the single KMN filling method, the LSTM predictor [46], the CNN network with residual connections [47], and the generative adversarial network (GAN) model [48] in the same test case. The interpolation results for randomly missing data are shown in Table 3.

It can be observed that as the probability of random missing data in PMU measurement increases, the imputation performance of the model decreases. KNN fills missing values by identifying and averaging the K nearest neighbors, without accounting for the topological relationships of each measurement point, resulting in poorer interpolation performance compared to KMN. These two simple methods, based on statistical calculations, do not perform as well as CNN and LSTM in extracting spatio-temporal features from measurement data, leading to larger imputation errors. Although GAN does not explicitly extract spatio-temporal features, its unique generation and comparison mechanism still results in considerable imputation performance. Overall, the proposed method achieves the smallest MAPE and MSPE values among the comparison models, delivering optimal performance. Thanks to the extraction and fusion of spatio-temporal features, the method, which explicitly models power grid topology, demonstrates more reliable imputation ability.

For the case of consecutive multi-point missing data, the test dataset mask is configured to apply a fixed-length block of missing data to each node's feature sequence, with an overall data missing rate of 30 %. As shown in the results of Table 4, as the number of consecutive missing points increases, the probability of missing neighboring measurement data for KNN and KMN also rises, making imputation more difficult and leading to larger test errors. In contrast, deep learning methods learn the overall data distribution, and even as the number of consecutive missing data points increases while maintaining the overall data missing rate, these models can still maintain good performance. The proposed method can extract high-order neighborhood features from the measurement data using the adjacency matrix, effectively capturing the spatial correlations within the system, and demonstrating better performance than other models.

Fig. 6 shows the model's performance when a PMU fault results in the loss of the entire measurement sequence, with 10 % corresponding to the missing measurement data from four nodes. It is evident that reconstructing the entire sequence through interpolation is more challenging than reconstructing randomly missing data. As the number of unobservable nodes increases, the model's reconstruction accuracy decreases to some extent, but the proposed method still outperforms the others.

4.3. Test results and comparison for TSA

When employing the proposed method for system stability evaluation, an accuracy of 81.16 % is achieved under a 30 % data loss rate in

Table 2
Model structure of the IEEE 39-bus system.

Input	Conv	Pooling + Readout	FC
39x12	(39x12,39x64) (20x64,20x64) (10x64,10x64)	(39x64,20x64) →(2x64) (20x64,10x64) →(2x64) (10x64, 5x64) →(2x64)	(128,64) (64,32) (32,2)

Table 3
Model performance under single point random missing.

Method	MAPE under data missing rate(%)			MSPE under data missing rate(%)		
	10 %	20 %	30 %	10 %	20 %	30 %
KNN	5.264	8.516	11.911	4.688	8.165	12.204
KMN	2.813	4.992	7.371	1.703	3.906	5.581
LSTM	0.378	0.732	1.204	0.064	0.096	1.147
CNN	0.325	0.671	1.104	0.055	0.089	1.129
GAN	0.294	0.618	1.008	0.033	0.081	0.112
Proposed	**0.254**	**0.530**	**0.881**	**0.024**	**0.077**	**0.091**

Table 4
Model performance under consecutive multi-point missing.

Method	MAPE under multi-point data missing(%)			MSPE under multi-point data missing (%)		
	3	4	5	3	4	5
KNN	15.120	19.997	25.273	13.871	17.226	22.505
KMN	10.023	11.702	13.975	8.146	9.304	11.068
LSTM	1.281	1.335	1.402	0.170	0.181	0.209
CNN	1.334	1.345	1.426	0.184	0.192	0.224
GAN	1.107	1.120	1.141	0.124	0.144	0.172
Proposed	**0.913**	**0.940**	**0.946**	**0.080**	**0.091**	**0.099**

the dataset. However, following the implementation of data reconstruction, the accuracy significantly improves to 98.28 %, demonstrating that the feature reconstruction module effectively enhances the performance of the downstream evaluation task.

In order to evaluate the actual performance of the model proposed, the following data-driven models are built based on the reconstructed data on the premise that the samples are exactly the same: SVM, LSTM, CNN, GCN, GAT, multi-scale GCN (MS-GCN), and multi-scale GAT (MS-GAT). The kernel function for SVM is the radial basis function, and the regularization parameter C is set to 10. The LSTM model consists of two LSTM layers and three fully connected layers, with the number of neurons in each layer being 256, 256, 64, 8, and 2, respectively. The CNN model includes two convolutional layers with a kernel size of 3x3, a pooling layer with a kernel size of 2x2, and a three-layer fully connected network. Both the GCN and GAT models consist of three layers for feature extraction, followed by a pooling layer and a three-layer fully connected network. The structure of the multi-scale GCN and multi-scale GAT models mirrors that of the proposed hierarchical graph pooling model, but with the GINConv layer replaced.

As shown in the test results in Table 5, the traditional machine learning method, SVM, has low evaluation accuracy, primarily due to its limited ability for manual feature extraction. In contrast, deep learning methods model the distribution of features in both space and time, which improves the stability judgment. The graph-based deep learning methods, GCN and GAT, capture the topological characteristics of the graph, while the hierarchical pooling structure effectively extracts and aggregates useful information from subgraphs at different scales. With this setup, the application of GINConv results in optimal performance across all evaluation metrics.

To validate the superiority of the proposed pooling method, comparative experiments are set up including the following methods: The non-pool model uses a three-layer GINConv as the feature extractor, which is directly connected to the classifier after feature extraction. The global-pooling method replaces the proposed pooling technique by summing or averaging all node features to obtain the graph's feature representation. The sag-pooling method sorts the results only based on the self-attention scores during the node selection process in the pooling layer. Additionally, to assess the effectiveness of the feature fusion mechanism, a variant is set up where the pooling layer discards the unselected nodes directly after the node selection process, without performing the first-order aggregation operation on the neighborhood.

It can be seen from Table 6 that proposed pooling method outperformed other methods in the test results. Graph pooling enhances feature expression by capturing global features of the entire graph,

Table 5
Comparison of different methods on IEEE 39-bus system.

Method	A_{cc}/%	R_{ec}/%	P_{re}/%	F_1/%
SVM	91.37	93.54	92.24	92.88
LSTM	95.81	96.92	96.92	96.92
CNN	94.18	95.93	95.53	95.73
GCN	95.64	98.56	95.93	96.83
GAT	95.78	97.68	96.08	96.69
MS-GCN	97.49	98.10	96.29	97.17
MS-GAT	97.71	98.12	96.81	97.52
Proposed	**98.28**	**98.74**	**98.22**	**98.47**

Table 6
Comparison of pooling methods on IEEE 39-bus system.

Method	A_{cc}/%	R_{ec}/%	P_{re}/%	F_1/%
Non-pooling	95.13	96.53	95.26	95.92
Global-pooling	97.15	97.80	97.55	97.77
Sag-pooling	97.49	98.41	97.76	98.08
W/o feature fusion	95.62	97.33	95.01	96.16
Proposed	**98.28**	**98.74**	**98.22**	**98.47**

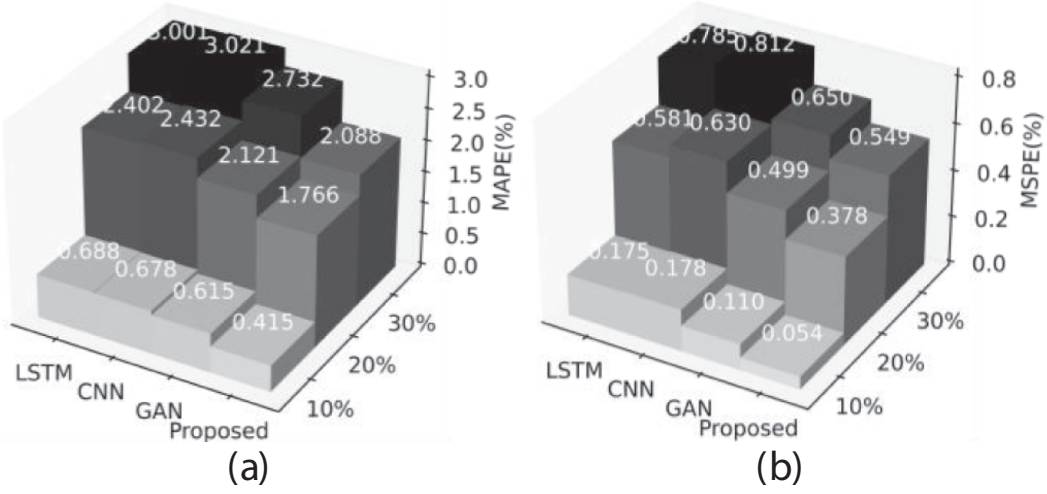


Fig. 6. Model performance under multiple sequences missing. (a) MAPE. (b) MSPE.

compared to non-pooling layers that solely use convolution operation. In contrast to simple global pooling that uses all node features, proposed pooling method considers differentiation of node features due to different topologies and location faults, leading to improved classification performance of the model. Compared to self-attention-based pooling methods, self-adaptive pooling fully considers the topology and timing characteristics of each node, resulting in more objective and reasonable node selection. Furthermore, node feature fusion is applied before discarding the unimportant nodes, ensuring that the node information is effectively utilized to obtain the subgraph representation.

To evaluate the classification ability of proposed model, t-SNE [49] method is employed to visualize the output of its middle layer. Fig. 7 displays the t-SNE visualization results of middle layer, with stable and unstable samples distinguished by different colors. The figure shows that the two sample types exhibit separability in the new feature space after feature extraction from the indistinguishable original data and are ultimately well differentiated after the fully connected layer.

To verify the model's adaptability to topological changes, 2585 samples are generated for testing by simulating three-phase short-circuit faults at various power flow levels and positions within the "N-2" topology. The parameters of all models remain unchanged, and the results are shown in Fig. 8. It can be observed that, although performance slightly declines, the proposed model consistently outperforms other methods in terms of both the degree of decline and overall accuracy. This is attributed to its ability to better capture the neighborhood characteristics of nodes in the graph and mitigate the impact of sample topology changes on the results, making it more robust than other data-driven methods.

4.4. Scalability test on a provincial power grid in China

4.4.1. Experimental setup and offline training

To validate the applicability of the model in real-world large-scale power systems, experiments are carried out based on a provincial power grid in China, illustrated in Fig. 9. Two typical power flow operation modes are considered, and three-phase short-circuit faults are applied to the head or tail of 500 kV AC lines, generating a total of 4588 samples, with 3676 stable and 1012 unstable cases. Given a sampling time interval of T = 0.01 s, each sample collected 10 consecutive PMU measurements for each bus after fault removal. Then the training set and test set are divided according to the ratio of 8: 2. To test the data reconstruction method and the robustness of the TSA model under noise interference and high data loss rates, the PMU failure rate is set to 10 %, and the data random loss rate is set to 30 % in the training set during offline training. For the test set, the PMU failure rate is increased to 20 %, the data random loss rate is raised to 40 %, and noise with varying signal-to-noise ratios is added. The parameter setting of the data

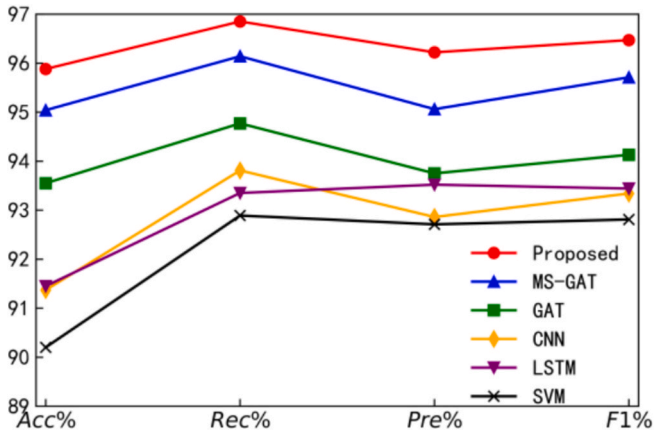


Fig. 8. Comparison of different methods on IEEE 39-bus system under "N-2" case.

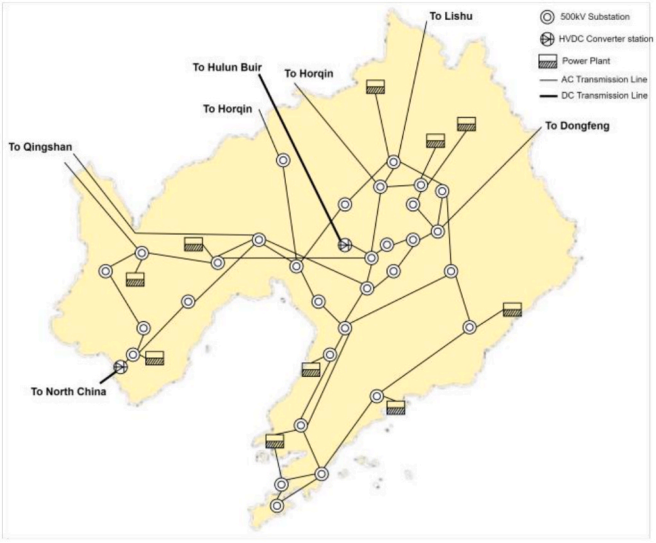


Fig. 9. Structure of a provincial power grid in China.

reconstruction module is the same as that described in section 1, and the hierarchical pooling TSA model is similar to the structure used in the IEEE 39 bus test system. The data interpolation results for the training and test sets are shown in Table 7. The interpolated training set data is used to train and validate the model's performance on the testing set. As

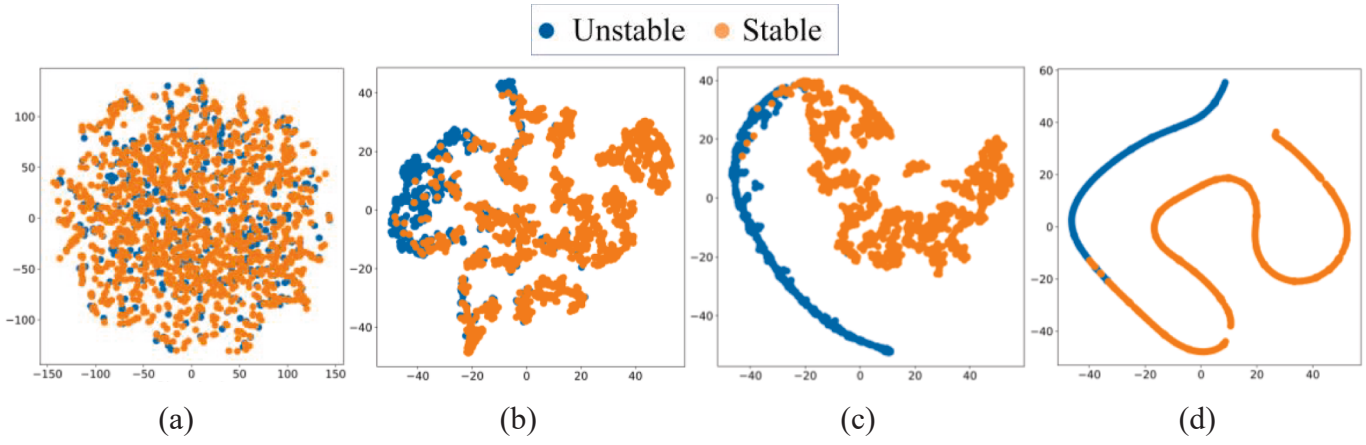


Fig. 7. T-SNE visualization of the proposed model. (a) original. (b) hierarchical pooling output. (c) fully connected layer I output. (d) final output.

shown in Table 8, the proposed model continues to outperform others on this power system.

4.4.2. Test results and online application

Table 9 presents the performance of data reconstruction and the TSA model under different levels of noise. The IEEE power system synchronous phasor data transmission standard specifies that the PMU phasor measurement error should not exceed 1 % [50]. At an equivalent measurement error of 1 %, the signal-to-noise ratio (SNR) is 20 dB. Based on test results, the model performance experiences a slight decrease at this noise level. Nevertheless, even under extreme conditions with a SNR ratio of 10 dB, the test accuracy remains above 97 %, demonstrating the model's strong robustness and its ability to meet the necessary requirements for online applications.

Time requirements also need to be considered when applying the model. Based on the reliability and security requirements of the power system, the transmission delay for PMU data should typically be kept within the range of tens of milliseconds to enable identification of transient events and pulse load changes in a timely manner. In addition to data acquisition and transmission delays, processing time and the time window for model input data must also be taken into account to provide more effective control time for emergency situations.

The following demonstrates an unstable case and the delay of the real-time model when applying trained model. A three-phase short-circuit grounding fault occurred at 5 % from the head end of a 500kv line at 1 s, and the maximum power angle difference of the whole system is monitored as shown in Fig. 10. The time (1) is the fault duration of 0.3 s. If no measures are applied, the maximum power angle difference of the system exceeds 360° at 1.77 s, and the system is unstable. Time (2) represents the action delay time, which includes the model input time window length of 100 ms, PMU transmission data delay of 50 ms, and data processing and model evaluation time of 10 ms. The TSA model evaluates that the system is unstable under this working condition and the generator shedding measure is applied at 1.46 s. Time (3) denotes the time difference between the application of stability control measures and the collapse of the system. Judging and applying measures before the system is unstable can ensure the stability of the system.

5. Conclusion

This paper proposes a hierarchical graph pooling method for power system TSA that accounts for missing data. Since electrical operation data in practical PMU applications may suffer from issues such as noise or missing values, it is crucial to recover this data quickly and accurately. Through comparative tests, the combination KNM interpolation and masked graph autoencoder reconstruction has been shown to outperform other methods under conditions of random data loss, consecutive multi-point missing, and PMU failure. This approach ensures the reliability and completeness of PMU measurement data, providing a solid data foundation for subsequent applications in the system. Specifically applied to power system TSA, the SAGpool mechanism is proposed to objectively evaluate the importance of nodes using both temporal and spatial features, thereby selecting nodes and integrating feature extraction from multi-scale subgraphs. The experimental verification and analysis were conducted using The IEEE 39-bus system and a provincial power grid in China. The model's accuracy reached 98.28 % and 99.24 %, while the F1 scores were 98.47 % and 99.86 %, respectively. In practical cases, where PMU measurement data serves as input, the model accounts for the influence of noise and the delay in model application. It is verified that the model can ensure timely stability judgments and facilitate the adoption of emergency control measures, thereby safeguarding the safe and stable operation of the power grid.

Future research will focus on integrating incremental learning, transfer learning, and other technologies to address the model's adaptability in real-world power grids, especially in larger and more complex systems. This will ensure the effectiveness of feature reconstruction methods and TSA models in previously unseen scenarios.

Table 7
Test results under different levels of noise.

Dataset	MAPE/%	MSPE/%
Training set	1.592	0.240
Testing set	1.668	0.251

Table 8
Comparison of different methods on a provincial grid in china.

Method	A_{cc}/%	R_{ec}/%	P_{re}/%	F_1/%
SVM	95.09	95.15	96.50	95.96
LSTM	97.61	99.03	97.81	98.41
CNN	97.52	98.73	98.36	98.54
GAT	97.71	98.21	98.89	98.55
MS-GAT	98.67	**99.85**	98.20	99.02
Proposed	**99.24**	99.72	**100**	**99.86**

Table 9
Test results under different levels of noise.

SNR /dB	MAPE/%	MSPE/%	A_{cc}/%	R_{ec}/%	P_{re}/%	F_1/%
30	1.683	0.265	98.70	99.08	99.26	99.17
20	1.886	0.387	98.27	98.53	99.26	98.89
10	2.112	0.486	97.24	97.55	98.09	97.82

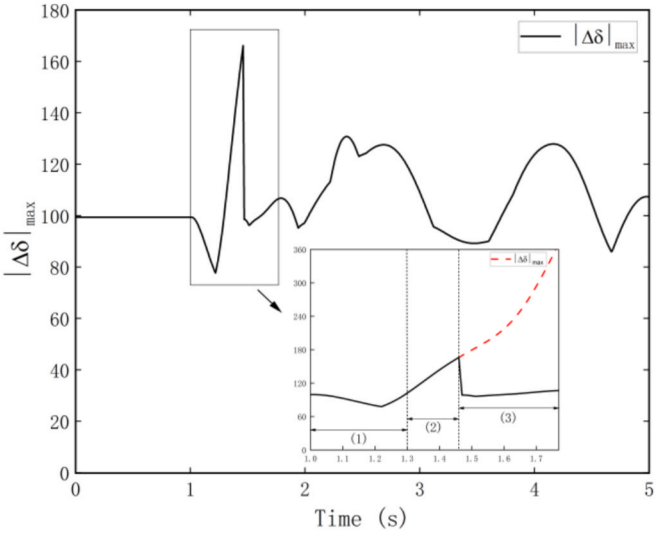


Fig. 10. The maximum power angle difference monitoring in an unstable case.

CRediT authorship contribution statement

Chenhao Zhao: Writing – original draft, Validation, Software, Methodology. **Zaibin Jiao:** Writing – review & editing, Supervision, Funding acquisition, Project administration. **Penghui Zhang:** Formal analysis, Data curation. **Linbo Zhang:** Software, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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